Lecture 1 — Notation, Functions, and Limits

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Georgetown University

8/18/25

Today's Agenda

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1. Introductions

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2. Math Camp Logistics and schedule

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3. Variables and measurement

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6. Series, sequences, and limits

Week's Agenda

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- Lecture 1: Notation, functions and limits

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- Lecture 2: Probability

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- Lecture 2: Probability
- Lecture 3: Calculus 1 - Derivatives
- Lecture 4: Calculus 2 - Integrals and multivariate calculus
- Lecture 5: Linear Algebra

Schedule

	Monday 18th	Tuesday 19th	Wednesday 20th	Thursday 21st	Friday 22nd
9:30 - 12:00	Notation, functions, and limits	Probability	Calculus 1	Calculus 2	Linear algebra
12:00 - 12:10	Break	Break	Break	Break	Break
12:10 - 1:00	Problem Set	Problem Set	R Basics	Problem Set	Intro to Tidyverse
1:00 - 2:00	Lunch Break	Lunch Break	Lunch Break	Lunch Break	Lunch Break
2:00 - 2:30	Review PS	Review PS	Review PS	Review PS	Intro to Tidyverse
2:30 - 3:30	Software Installation	Intro to R	Analysis in R	Intro to Stata	Q&A

Goals of Math Camp

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- We use quantitative methods to test theories regarding the relationships between concepts
- The goal of a "causal claim" is rarely achieved in full, but we can make compelling arguments
- Potential Outcomes Framework (POF) is one approach to pursue causal inference
- Qualitative methods can answer "how" and "why" questions

Quantitative vs. Qualitative Methods:

Table 1 Contrasting qualitative and quantitative research

Section	*Criterion*	*Qualitative*	*Quantitative*
1	Approaches to explanation	Explain individual cases; "causes-of-effects" approach	Estimate average effect of independent variables; "effects-of-causes" approach
2	Conceptions of causation	Necessary and sufficient causes; mathematical logic	Correlational causes; probability/statistical theory
3	Multivariate explanations	INUS causation; occasional individual effects	Additive causation; occasional interaction terms
4	Equifinality	Core concept; few causal paths	Absent concept; implicitly large number of causal paths
5	Scope and generalization	Adopt a narrow scope to avoid causal heterogeneity	Adopt a broad scope to maximize statistical leverage and generalization
6	Case selection practices	Oriented toward positive cases on dependent variable; no (0,0,0) cases	Random selection (ideally) on independent variables; all cases analyzed
7	Weighting observations	Theory evaluation sensitive to individual observations; one misfit can have an important impact	All observations are a priori equally important; overall pattern of fit is crucial
8	Substantively important cases	Substantively important cases must be explained	Substantively important cases not given special attention
9	Lack of fit	Nonconforming cases are examined closely and explained	Nonsystematic causal factors are treated as error
10	Concepts and measurement	Concepts center of attention; error leads to concept revision	Measurement and indicators center of attention; error is modeled and/or new indicators identified

229

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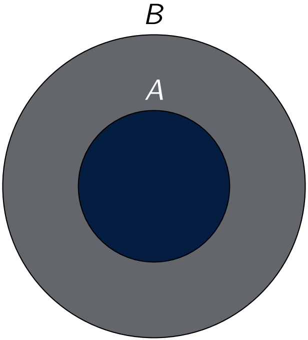
- **Mutually exclusive sets:** the intersection is the empty set.

$A \subset B$ (Proper Subset)



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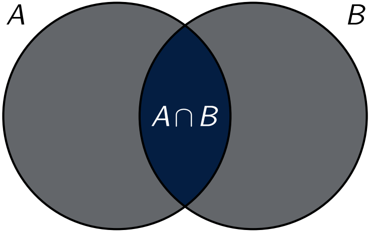

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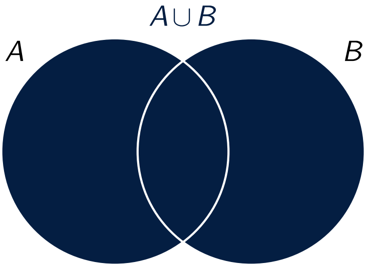
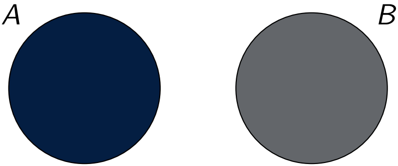


$A \cap B = \emptyset$ (Mutually Exclusive)



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    - Ratio: meaningful zero (re-scaled thermometer)

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  - Categorical: pie charts, bar charts
  - Numerical (especially Continuous): histograms, box plots
- Analysis
  - Numerical data can use summary statistics (mean, median, standard deviation)
  - Categorical - Categorical : cross-tabulation
  - Categorical - Continuous : difference in means
  - Continuous - Continuous : scatter plots, regression

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- Roots: $\sqrt[a]{x}$
  - All roots can be expressed as fractional exponents
  - $\sqrt[a]{x} = x^{1/a}$
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- Natural log: log base e
  - $e \approx 2.7183$
  - Useful properties in calculus
  - Taking the natural log of a variable allows us to interpret relationships in terms of percentage changes

Exponent, logarithms and root rules

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- $log(x^b) = b \cdot ln(x)$ for $x > 0$

Exponent, logarithms and root rules

Root rules

- $\sqrt{x} = x^{\frac{1}{2}}$
- $\sqrt[n]{x} \cdot \sqrt[n]{z} = \sqrt[n]{xz}$
  for $n > 1$
- $\frac{\sqrt[n]{x}}{\sqrt[n]{z}} = \sqrt[n]{\frac{x}{z}}$
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- B is the *codomain*, or set of possible y values.
- To be a function, there must only be one unique value in its range (y) for each value in its domain (x)

Graph examples

In which of these graphs can we observe a function?

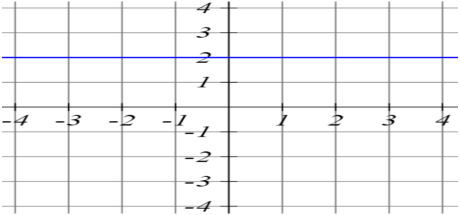


Figure: a)

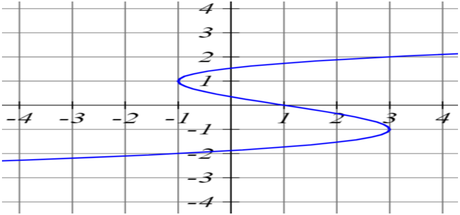


Figure: b)

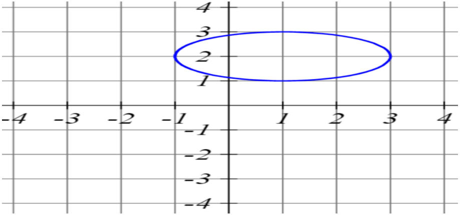
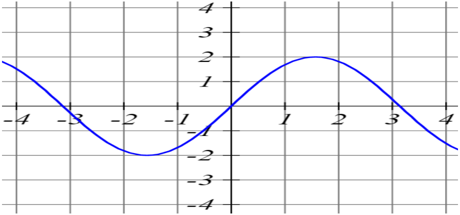


Figure: c)



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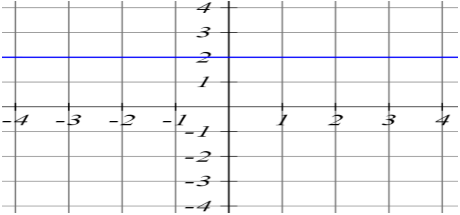

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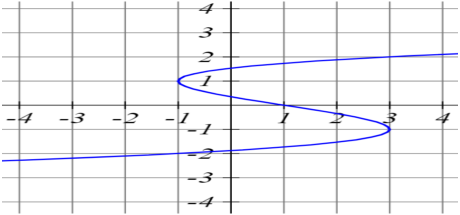

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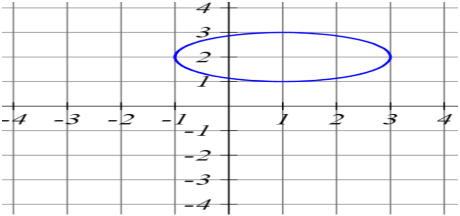

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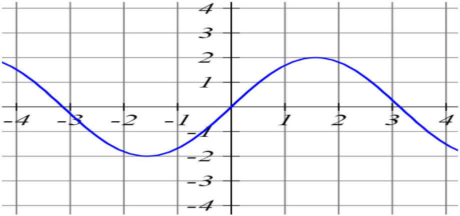

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Examples of functions of one variable - linear equation

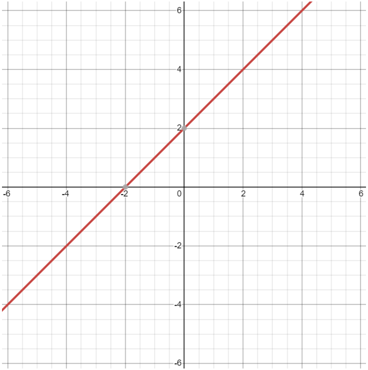

Figure: y=2+x

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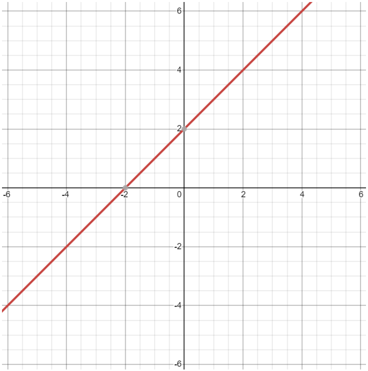

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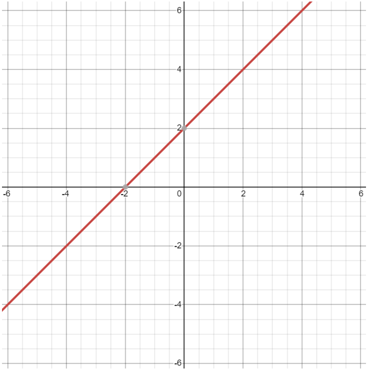

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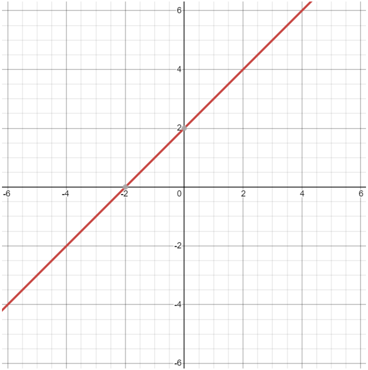

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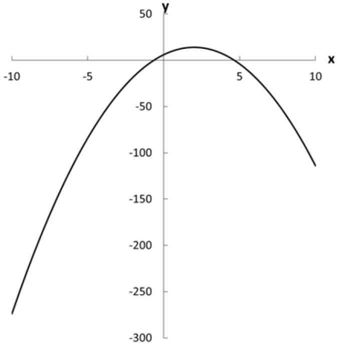
Examples of functions of one variable - Quadratic function



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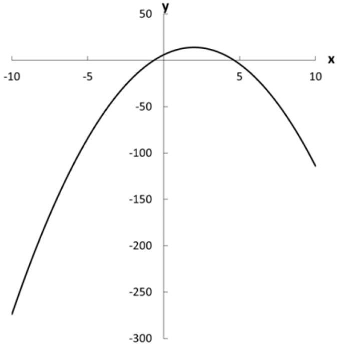

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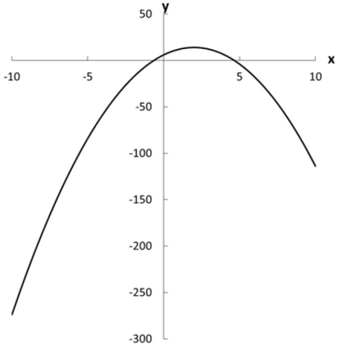

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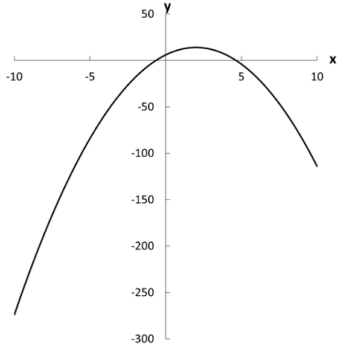

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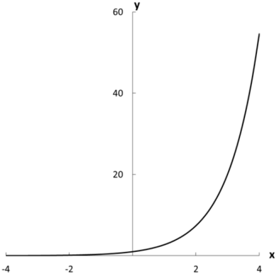
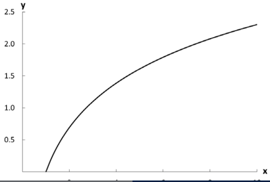
Exponent, logarithms and roots - Graphs

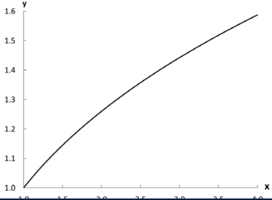
Exponential function



Logarithmic function



Root function



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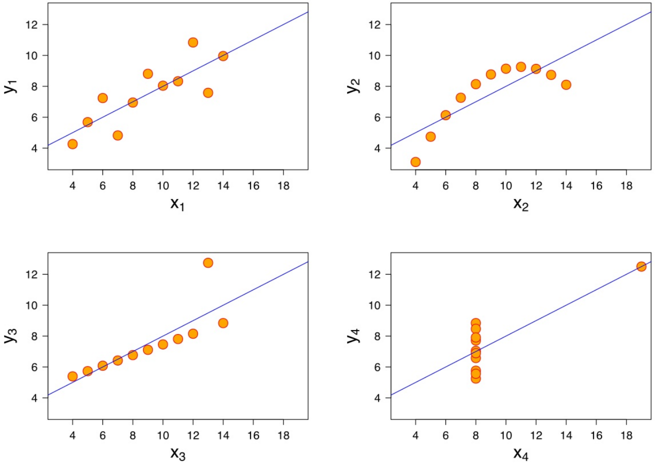
- Most quantitative social science boils down to how two variables relate!
- Our examples suggest a precise function between x and y
- In statistics, you will *theorize* the functional form of the relationship
- Statistical methods will *estimate* the most appropriate parameters; "a line of best fit"

Anscombe's Quartet



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$$\frac{x(x-1)}{2} \tag{5}$$

Quadratics and Special Products

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$$x = 1 \text{ OR } x = 7$$

Why two solutions?

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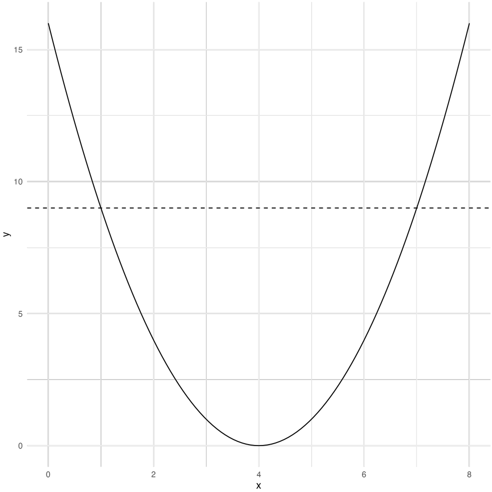

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Either x + 3 = 0 OR x + 4 = 0

$x = -3$ OR $x = -4$

Quadratic Formula

Useful if special products can't easily be found!

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{15}$$

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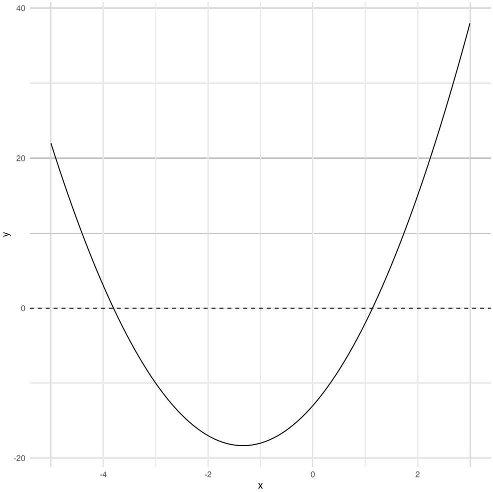
$$x \approx 1.14 \tag{19}$$

$$x = \frac{-8 - \sqrt{220}}{6} \tag{20}$$

$$x \approx -3.81 \tag{21}$$

Checking our work graphically



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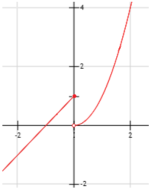
- Piecewise function: the function (relationship between x and y) is different for different parts of the domain

Piecewise Function Example

Example: Estimate the value of the following limits: $\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0^+} f(x)$ for the following function:



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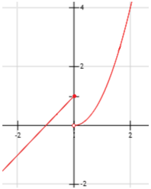
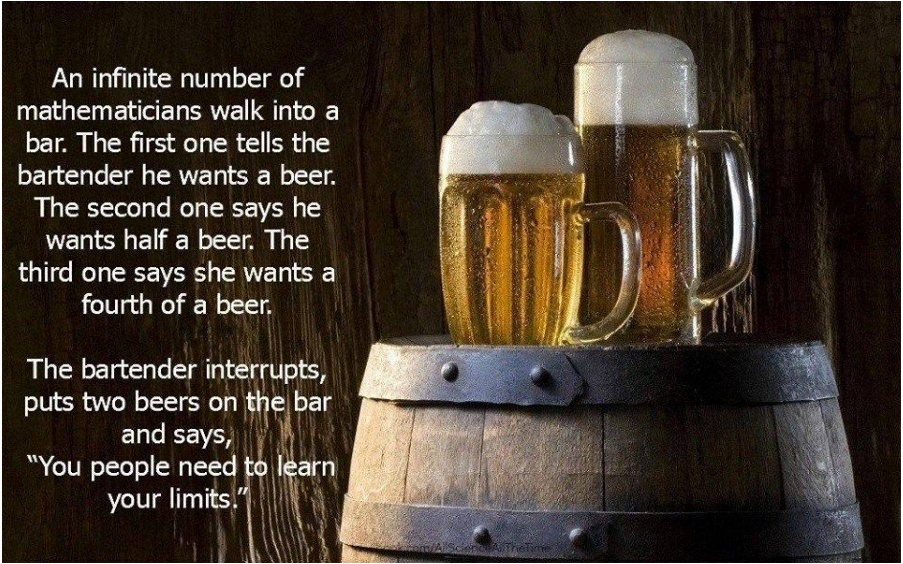


$$f(x) = \begin{cases} x+1 & x \le 0 \\ x^2 & x > 0 \end{cases}$$

$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} x+1 = 1$$

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} x^2 = 0$$


An infinite number of mathematicians walk into a bar. The first one tells the bartender he wants a beer. The second one says he wants half a beer. The third one says she wants a fourth of a beer.
The bartender interrupts, puts two beers on the bar and says,
"You people need to learn your limits."

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- Functions are **continuous** if they do not have sudden breaks, and **discontinuous** otherwise

Math Camp Exercises - Day 1

1. M&S Chapter 1: Exercises 3, 5
2. M&S Chapter 2: Exercises 16, 17, 19, 20, 21, 22, 26, 29
3. M&S Chapter 3: Exercises 2, 3
4. M&S Chapter 4: Exercises 3, 4, 5, 6, 9